Sphingobacterium multivorum DSM 11691

Isolate Assembly QC Report 05/10/2014

1. Project Information

Program	Microbial/CSP 2012
PMO Project	0
Seq Proj ID	1030929
Sequencing Project Name	Sphingobacterium multivorum DSM 11691
JGI Project ID	0

2. Read Statistics

Illumina Std PE Statistics

File name	7779.1.83549.GAATAA.fastq
Library	TSWG
Number of reads	14,341,954
Sequencing depth †	430X
Read type	2x150 bp

† A genome size of 5.0 Mbp was assumed in this calculation.

3. Read QC Results

The following are the results of reads screened against contaminants. Pairs of matching reads were removed from the dataset.

Illumina Std PE Read Filter Statistics

Description	Num Reads	Pct Reads
Input	14,341,954	100
Artifact removed	68,542	0.5
Total removed	4,341,954	30.3
Total remaining	10,000,000	69.7

The following are the results of reads screened against potential reagent and process contaminants but were not removed from the dataset.

Illumina Std PE Contamination Identification Statistics

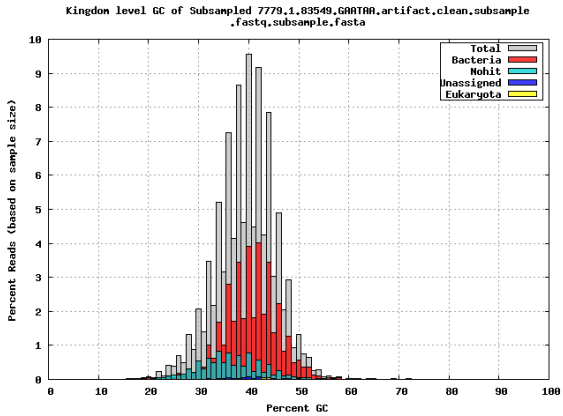
Description	Num Reads	Pct Reads
Input	14,341,954	100
Contam identified	4	0.0

List of Contaminants Identified

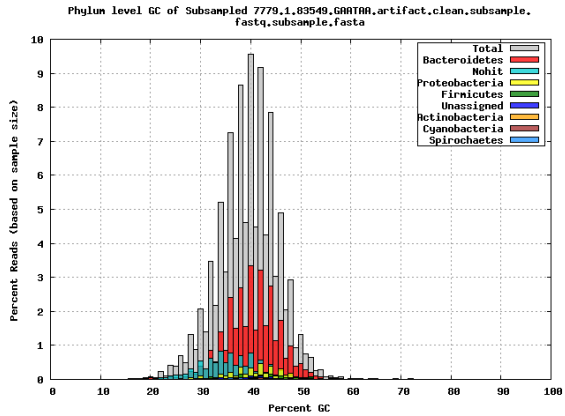
Description	Num Reads	Pct Reads
Escherichia	2	0.00
Ralstonia	2	0.00

GC histogram of the reads subsampled to 10k, overlaid with GC of hits based on BLASTX, shown for different taxonomic levels.

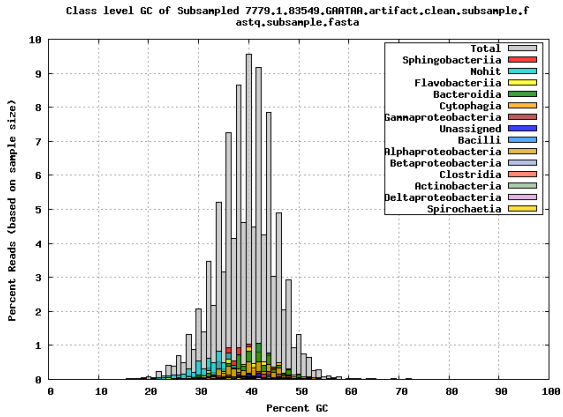
Kingdom



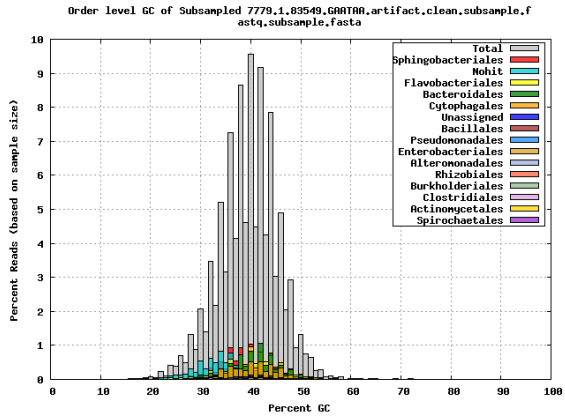
Phylum



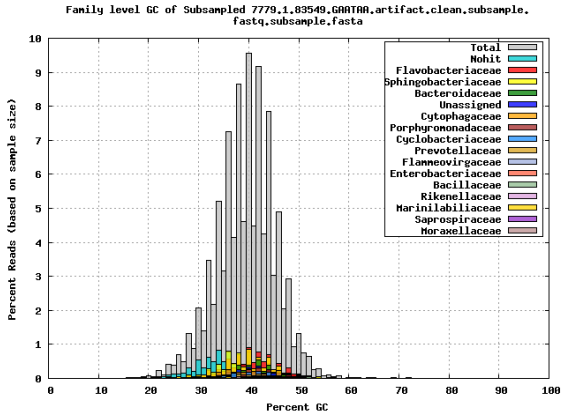
Class



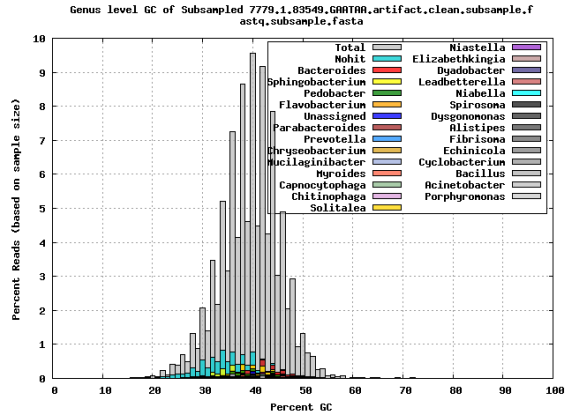
Order



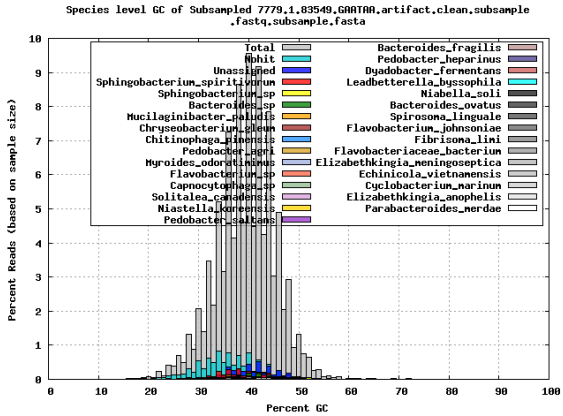
Family



Genus



Species



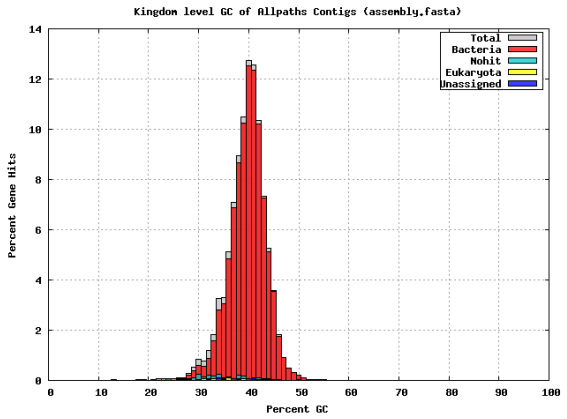
4. Assembly Statistics

Assembly method	Allpaths/Velvet
Scaffold total	88
Contig total	90
Scaffold sequence length	5.9 Mb
Contig sequence length	5.9 Mb (0.0% gap)
Scaffold N/L50	16/112.1 kb
Contig N/L50	17/111.1 kb
Largest Contig	381.3 kb
Number of scaffolds >50 kb	42
Pct of genome in scaffolds >50 kb	85.6

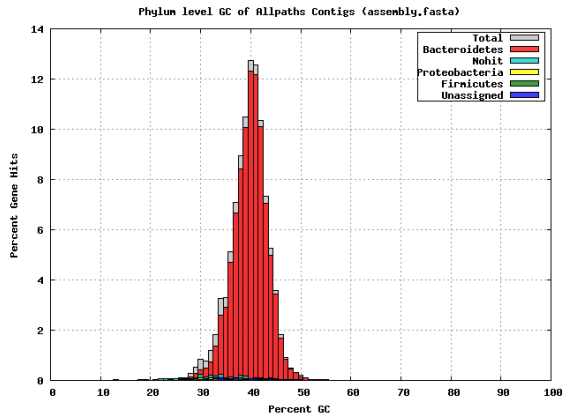
5. Assembly QC Results

GC histogram of the predicted genes on each contig, overlaid with GC of hits based on BLASTP, shown for different taxonomic levels.

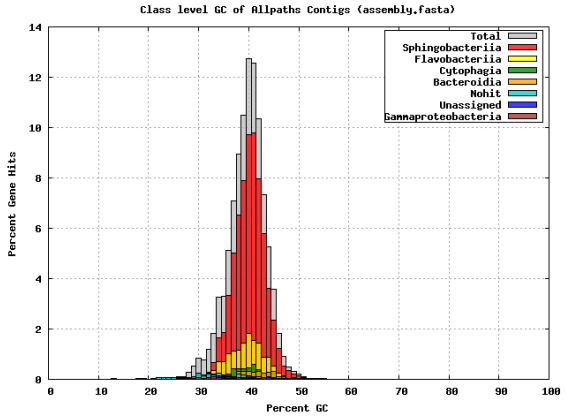
Kingdom



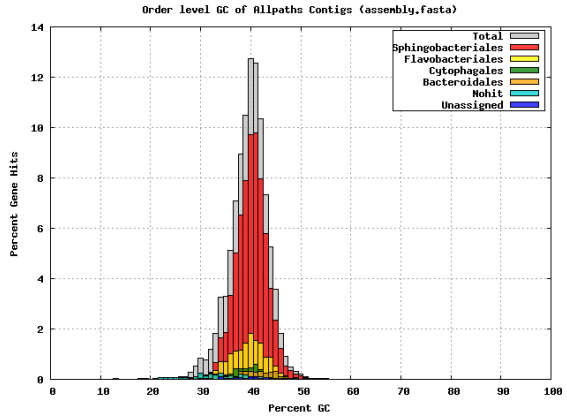
Phylum



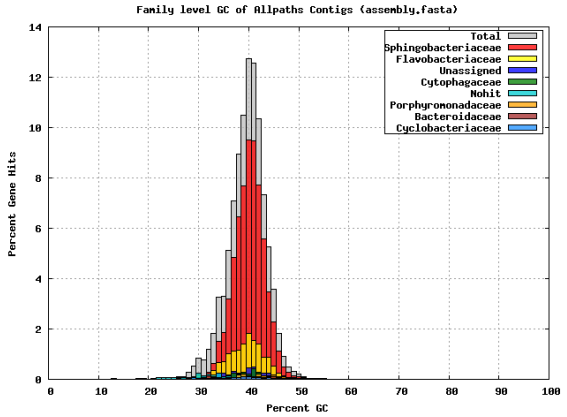
Class



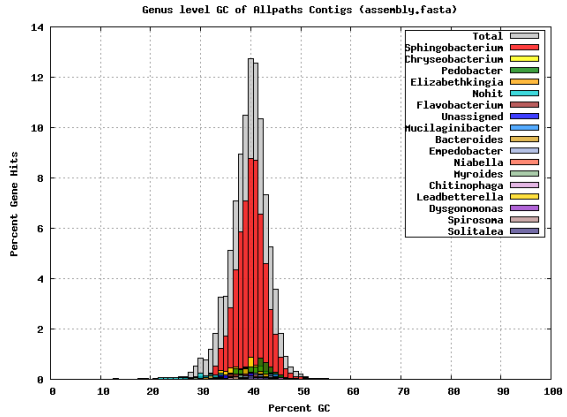
Order



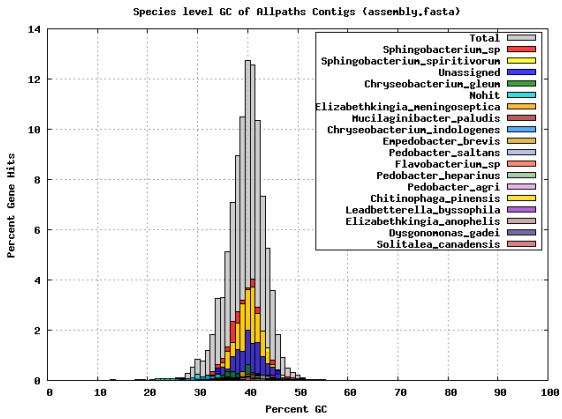
Family



Genus

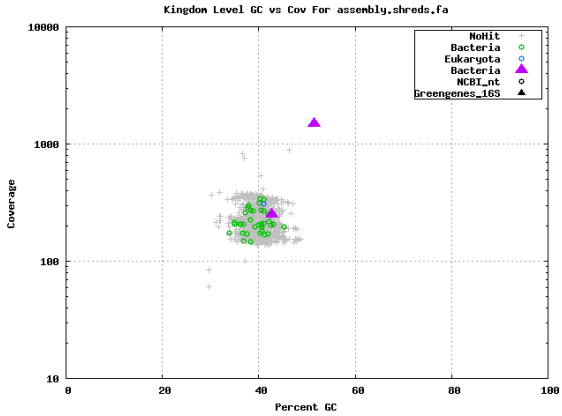


Species

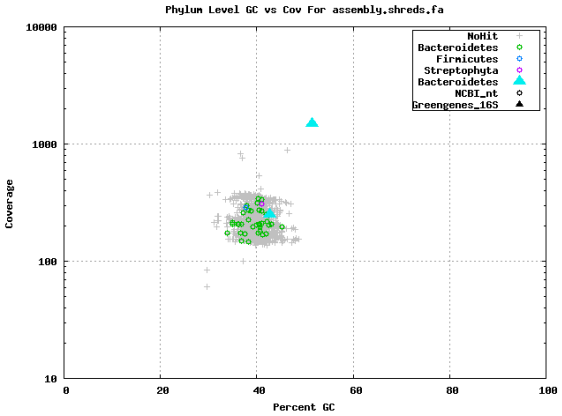


GC vs coverage based on GC of NCBI nt and Greengenes 16S rRNA gene hits to the assembly using megablast, shown for different taxonomic levels.

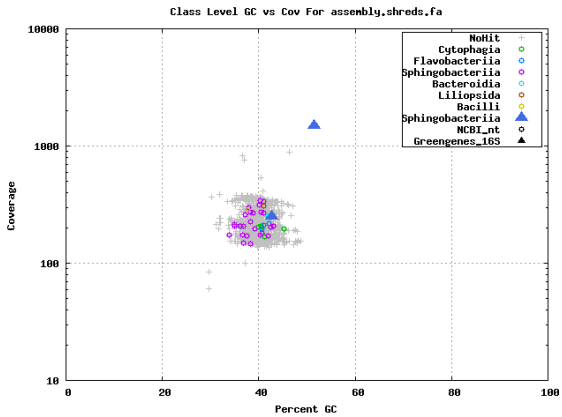
Kingdom



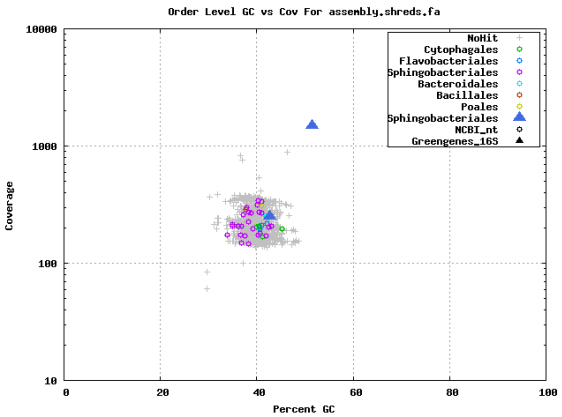
Phylum



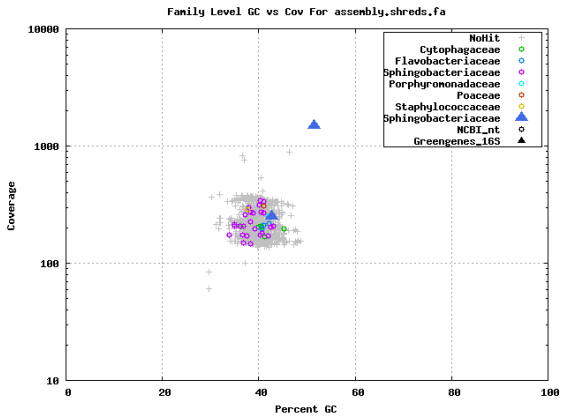
Class



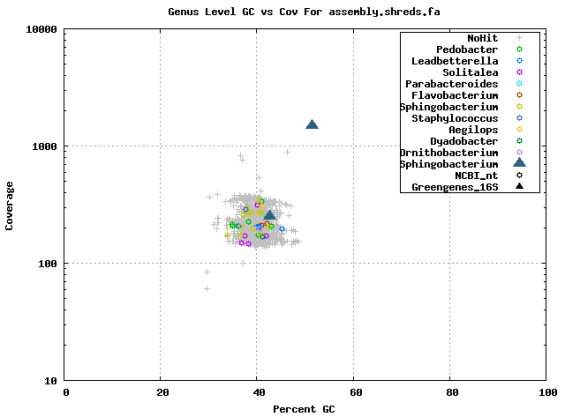
Order



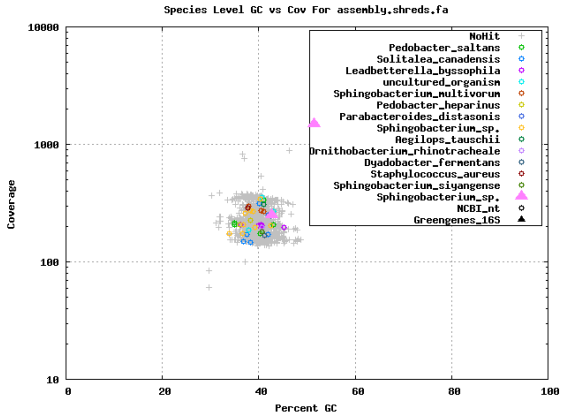
Family



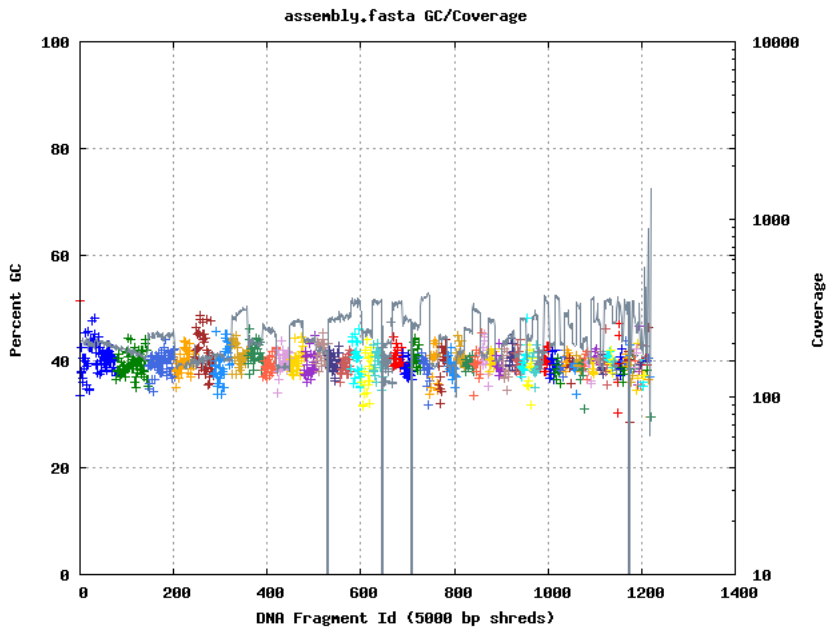
Genus



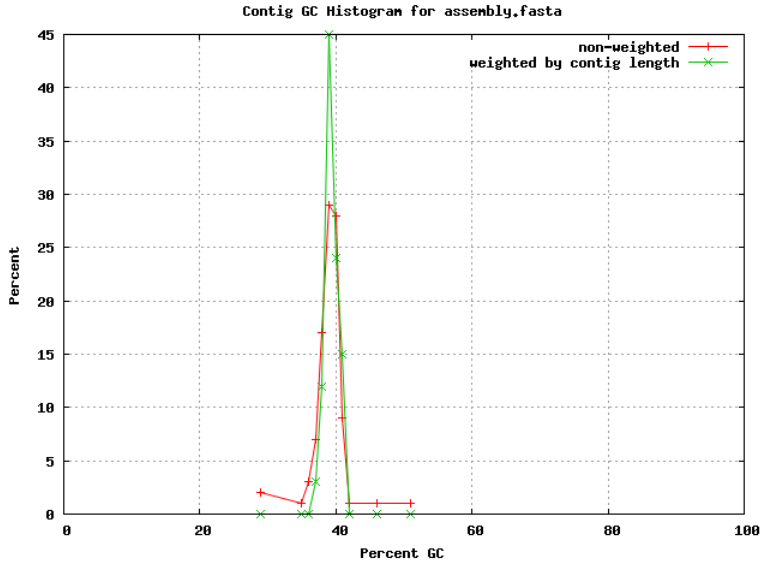
Species



Coverage vs GC. Contigs were shredded into non-overlapping 5kbp and the GC of each shred was plotted as a point, colored by scaffold id. Coverage was calculated by mapping the fragment library to the final asssembly and plotted as connected points.



GC histogram of the contigs, including contig length weighted distribution.

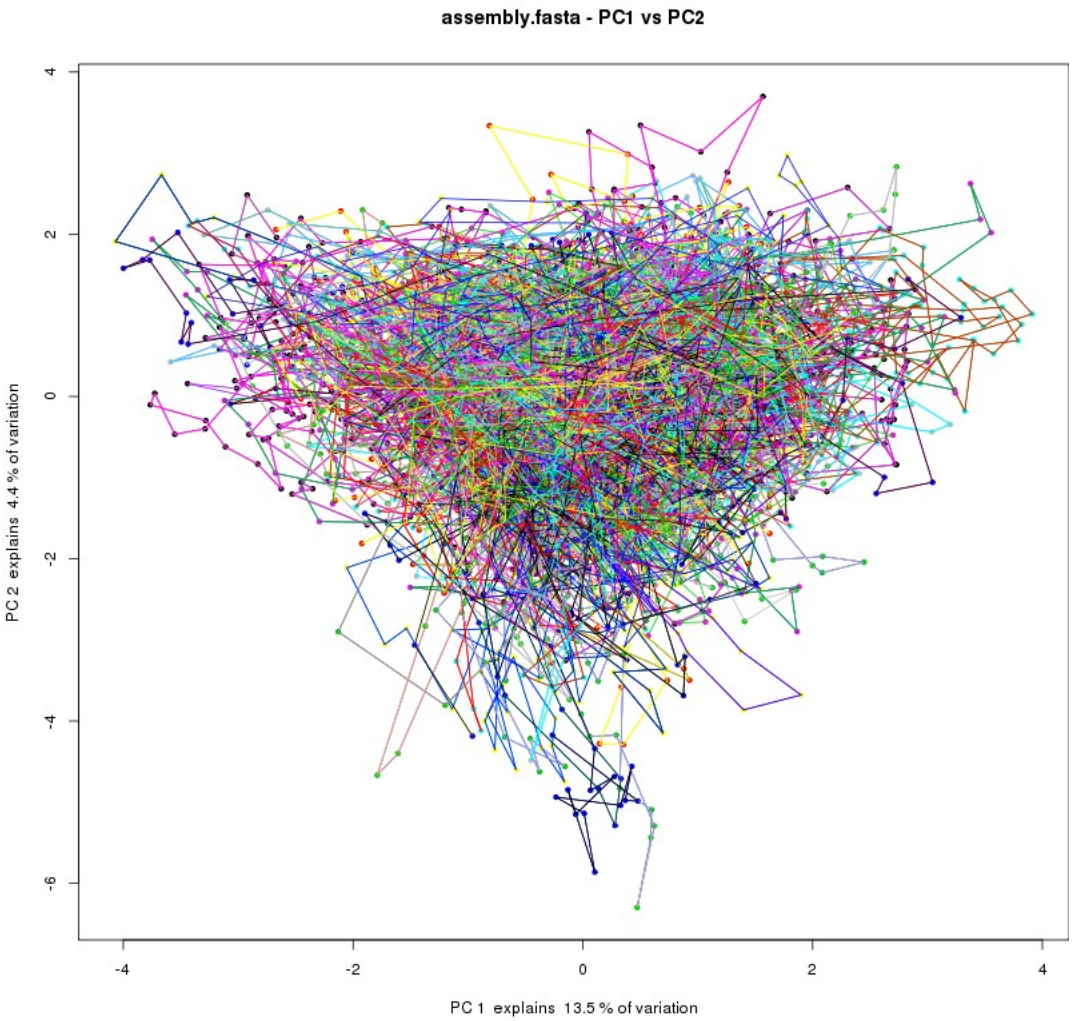


List of contigs and average percent GC, grouped in bins of 5:

Pct GC Bin	Contig Name
25	scaffold00087, scaffold00088
35	scaffold00002, scaffold00003, scaffold00004, scaffold00006, scaffold00009, scaffold00010, scaffold00012, scaffold00014, scaffold00015, scaffold00017, scaffold00018, scaffold00019, scaffold00021, scaffold00023, scaffold00024, scaffold00025, scaffold00026, scaffold00028, scaffold00029, scaffold00032, scaffold00033, scaffold00036, scaffold00037, scaffold00038, scaffold00041, scaffold00042, scaffold00043, scaffold00045, scaffold00046, scaffold00047, scaffold00048, scaffold00050, scaffold00051, scaffold00054, scaffold00057, scaffold00058, scaffold00059, scaffold00060, scaffold00061, scaffold00062, scaffold00067, scaffold00071, scaffold00072, scaffold00073, scaffold00074, scaffold00075, scaffold00076, scaffold00077, scaffold00082, scaffold00084, scaffold00086

40	scaffold00001, scaffold00005, scaffold00007, scaffold00008, scaffold00011, scaffold00013, scaffold00016, scaffold00020, scaffold00022, scaffold00027, scaffold00030, scaffold00031, scaffold00034, scaffold00035, scaffold00039, scaffold00040, scaffold00044, scaffold00049, scaffold00052, scaffold00053, scaffold00055, scaffold00056, scaffold00063, scaffold00064, scaffold00065, scaffold00066, scaffold00068, scaffold00069, scaffold00070, scaffold00078, scaffold00079, scaffold00080, scaffold00081 scaffold00083
45	scaffold00085
50	NODE_1_length_1470_cov_111.012245

Principal component analysis of tetramer frequencies of contigs. Detectable variations are highlighted in color.



6. Sequence Data Availability

The following sequence fasta files can be downloaded from our JGI portal website.
http://www.jgi.doe.gov/genome-projects

Filename	Description
submission.assembly.fasta	Allpaths/Velvet

7. Annotation Data Availiability

The annotation of the assembled contigs can be found within IMG.
http://img.jgi.doe.gov

8. Methods

Isolate Minimal Draft

Genome sequencing and assembly
The draft genome of was generated at the DOE Joint genome Institute (JGI) using the Illumina technology [1]. An Illumina std shotgun library was constructed and sequenced using the Illumina HiSeq 2000 platform which generated 14,341,954 reads totaling 2,151.3 Mb. All general aspects of library construction and sequencing performed at the JGI can be found at http://www.jgi.doe.gov. All raw Illumina sequence data was passed through DUK, a filtering program developed at JGI, which removes known Illumina sequencing and library preparation artifacts [2]. Following steps were then performed for assembly: (1) filtered Illumina reads were assembled using Velvet (version 1.2.07) [3], (2) 1–3 kb simulated paired end reads were created from Velvet contigs using wgsim [4], (3) Illumina reads were assembled with simulated read pairs using Allpaths–LG (version r46652) [5]. Parameters for assembly steps were: 1) Velvet (velveth: 63 –shortPaired and velvetg: –very_clean yes –exportFiltered yes –min_contig_lgth 500 –scaffolding no –cov_cutoff 10) 2) wgsim (–e 0 –1 100 –2 100 –r 0 –R 0 –X 0) 3) Allpaths–LG (PrepareAllpathsInputs: PHRED_64=0 PLOIDY=1 FRAG_COVERAGE=125 JUMP_COVERAGE=25 LONG_JUMP_COV=50, RunAllpathsLG: THREADS=8 RUN=std_shredpairs TARGETS=standard VAPI_WARN_ONLY=True OVERWRITE=True). The final draft assembly contained 90 contigs in 88 scaffolds, totalling 5.9 Mb in size. The final assembly was based on 1,500.0 Mb of Illumina data. 300.0X input read coverage was used for the final assembly.

Genome annotation
Genes were identified using Prodigal [6], followed by a round of manual curation using GenePRIMP [7] for finished genomes and Draft genomes in fewer than 20 scaffolds. The predicted CDSs were translated and used to search the National Center for Biotechnology Information (NCBI) nonredundant database, UniProt, TIGRFam, Pfam, KEGG, COG, and InterPro databases. The tRNAScanSE tool [8] was used to find tRNA genes, whereas ribosomal RNA genes were found by searches against models of the ribosomal RNA genes built from SILVA [9]. Other non–coding RNAs such as the RNA components of the protein secretion complex and the RNase P were identified by searching the genome for the corresponding Rfam profiles using INFERNAL [10]. Additional gene prediction analysis and manual functional annotation was performed within the Integrated Microbial Genomes (IMG) platform [11] developed by the Joint Genome Institute, Walnut Creek, CA, USA [12].

1. Bennett S. Solexa Ltd. Pharmacogenomics. 2004;5(4):433–8.
2. Mingkun L, Copeland A, Han J. DUK, unpublished, 2011.
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4. https://github.com/lh3/wgsim.
5. Gnerre S, MacCallum I. High–quality draft assemblies of mammalian genomes from massively parallel sequence data. PNAS 2011;108:4 1513–1518.
6. Hyatt D, Chen GL, Lacascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. BMC Bioinformatics 2010; 11:119.
7. Pati A, Ivanova NN, Mikhailova N, Ovchinnikova G, Hooper SD, Lykidis A, Kyrpides NC. GenePRIMP: a gene prediction improvement pipeline for prokaryotic genomes. Nat Methods 2010; 7:455–457.
8. Lowe TM, Eddy SR. tRNAscan–SE: a program for improved detection of transfer RNA genes in genomic sequence. Nucleic Acids Res 1997; 25:955–964.
9. Pruesse E, Quast C, Knittel, Fuchs B, Ludwig W, Peplies J, Glckner FO. SILVA: a comprehensive online resource for quality checked and aligned ribosomal RNA sequence data compatible with ARB. Nuc Acids Res 2007; 35: 2188–7196.
10. INFERNAL. Inference of RNA alignments. http://infernal.janelia.org.
11. The Integrated Microbial Genomes (IMG) platform. http://www.ncbi.nlm.nih.gov/pubmed/24165883

12. Markowitz VM, Mavromatis K, Ivanova NN, Chen IMA, Chu K, Kyrpides NC. IMG ER: a system for microbial genome annotation expert review and curation. Bioinformatics 2009; 25:2271–2278.